



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
03-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

Time : 3 Hours

Maximum Marks : 222

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **33** pages and that all the **22** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

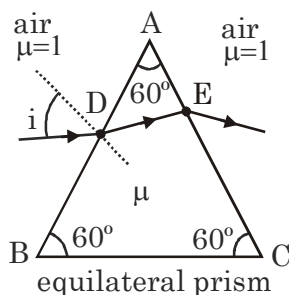
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Zero Marks : 0 In all other cases.

1. A light ray is incident on equilateral prism of side length d on mid point at variable angle of incidence i . Now choose the correct answer. [D is mid point of side AB]



- (A) If $\mu = \sqrt{2}$, minimum value of distance AE is (without any TIR from side AC) $\frac{d}{2}(\sqrt{3} - 1)$.
- (B) If $\mu = \sqrt{2}$, angle of minimum deviation produced by prism is 30° .
- (C) If $\mu > 2$ then TIR take place for all value of i .
- (D) In case of minimum deviation emergent ray must be emerging from midpoint of face AC.

Space for Rough Work

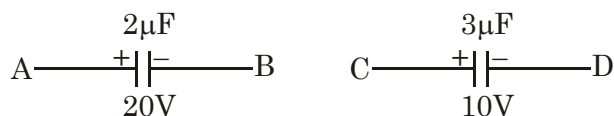
2. Minimum wavelength of light emitted by excited positronium atom is λ (Positronium is formed by an electron and a positron) $R = \text{Rydberg constant}$.

(A) $\lambda > \frac{3}{R}$ (B) $\lambda > \frac{1.5}{R}$ (C) $\lambda > \frac{1}{R}$ (D) $\lambda > \frac{2.5}{R}$

3. A charged particle having charge 'Q' and mass 'm' is released from rest in a region where mutually perpendicular electric field (E) and magnetic field (B) is present. What could be possible speed of particle during subsequent motion ?

(A) $1.5 \frac{E}{B}$ (B) $2.5 \frac{E}{B}$ (C) $1.7 \frac{E}{B}$ (D) $1.9 \frac{E}{B}$

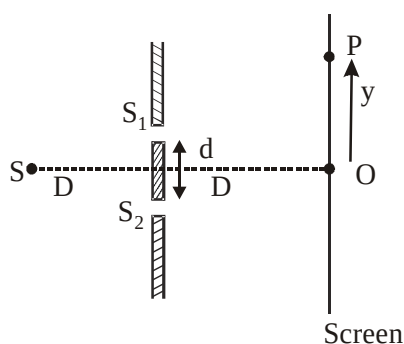
4. In the given figure if A is connected with C and B is connected with D, then



- (A) Charge flows in the circuit is $12 \mu\text{C}$. (B) Heat produced in the circuit is $60 \mu\text{J}$.
 (C) Charge flows in the circuit is $14 \mu\text{C}$. (D) Heat produced in the circuit is $490 \mu\text{J}$.

Space for Rough Work

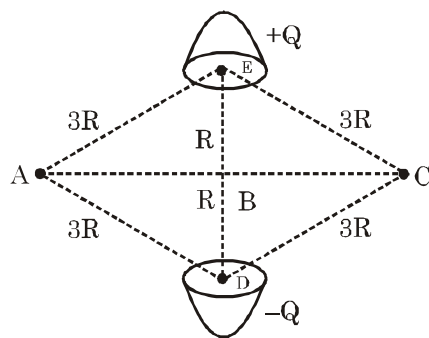
5. Pick the **CORRECT** statement/statements from the following :
- (A) Viscosity in liquids, is the property which is responsible for transport of momentum from one layer of liquid to other.
- (B) Viscous resistance force in fluids and kinetic friction force in solids are non-conservative forces.
- (C) Ideally, a solid steel sphere, starting from rest, requires infinite time to acquire terminal velocity in a big reservoir of honey.
- (D) In liquids, the cause of viscosity, is the cohesive force.
6. Refer to YDSE setup in given figure, a monochromatic (wavelength λ) coherent source S is present, interference is observed at point P. S_1 & S_2 are slits of equal small width, at separation d . Source and screen are both at distance D from mid point of slits S_1 & S_2 . O is symmetric from S_1 & S_2 .



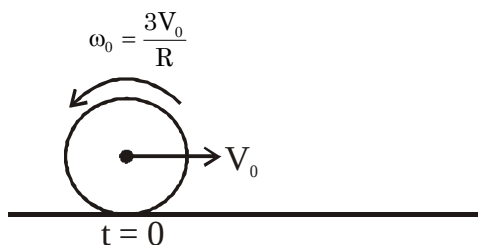
- (A) For $d = 1\text{mm}$, $D = 0.5\text{m}$, $\lambda = 1\mu\text{m}$, at $y = 0.25\text{m}$, 500^{th} maxima is formed.
- (B) For $d = \sqrt{21}\text{cm}$, $D = 10\text{cm}$, $\lambda = 1\text{cm}$, in front of S_1 & S_2 there is first minima.
- (C) For $d = \sqrt{21}\text{cm}$, $D = 10\text{cm}$, $\lambda = 1\text{cm}$, in front of S_1 & S_2 there is first maxima.
- (D) For $d = 1\text{mm}$, $D = 0.5\text{m}$, $\lambda = 1\mu\text{m}$, at $y = 5\text{mm}$, 10^{th} maxima is formed

Space for Rough Work

7. Charge $+Q$ and $-Q$ are distributed uniformly on two hemispheres of radius R as shown, then



- (A) Potential at points A and C are same. (B) Potential at points A, B and C are same
(C) Potential at E and D are same (D) None of these
8. A uniform disc of radius R moves on a rough horizontal surface with initial velocity V_0 and angular velocity ω_0 as shown. Coefficient of friction is μ . Which of the following is/are correct?



- (A) Time t_1 when pure rolling starts is $\frac{4V_0}{3\mu g}$.
- (B) Time taken by disc to come back to original position is $\frac{8V_0}{3\mu g}$.
- (C) Disc will never come back to original position.
- (D) Total work done on the disc is negative.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Four or Five or Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Consider a diatomic gas enclosed in a cylinder with a movable piston. The piston is massless, and the pressure outside is atmospheric pressure. If the cylinder is kept on a heat reservoir of constant temperature T and the piston is pulled out slowly so that the temperature of the gas inside remains constant, then

List-I

- (I) Remains constant
 (II) Increases
 (III) Decreases
 (IV) Cannot be predicted with data given

List-II

- (P) The force required to pull out the piston
 (Q) The average change in momentum of molecules colliding with the piston
 (R) The number of collision of molecules per unit time with the walls
 (S) Molar mass of the gas
 (T) The pressure of the gas
 (U) The average distance between molecules

(A) I $\rightarrow$ Q; II $\rightarrow$ R,U; III $\rightarrow$ S,T; IV $\rightarrow$ P,U

(B) I $\rightarrow$ P,U; II $\rightarrow$ P,U; III $\rightarrow$ R,S,T; IV $\rightarrow$ S

(C) I $\rightarrow$ Q,S; II $\rightarrow$ P,U; III $\rightarrow$ R,T; IV $\rightarrow$ S

(D) I $\rightarrow$ Q,S; II $\rightarrow$ R,U; III $\rightarrow$ P,T,U; IV $\rightarrow$ S

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Consider a diatomic gas enclosed in a cylinder with a movable piston. The piston is massless, and the pressure outside is atmospheric pressure. If the piston as well as the cylinder are adiabatic, and the gas undergoes adiabatic expansion such that volume increases slowly.

List-I

- (I) Remains constant
- (II) Increases
- (III) Decreases
- (IV) Cannot be predicted with data given

List-II

- (P) Molar mass of the gas
- (Q) The average change in momentum of molecules colliding with the piston
- (R) The number of collision of molecules per unit time with the walls
- (S) The rate of flow of heat energy into the gas
- (T) The pressure of the gas
- (U) The average distance between molecules

(A) I → P; II → U; III → R,S; IV → S,T

(B) I → S; II → U; III → Q,R,T; IV → P

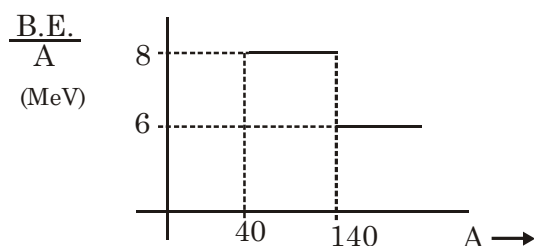
(C) I → P; II → U; III → Q,S,T; IV → P

(D) I → S; II → U; III → Q,T; IV → P

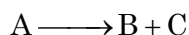
Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Hypothetical B.E. per nucleon versus mass number is given



Consider nuclear Fission reaction.



List I gives different reaction and List II gives energy in MeV.

List-I	List-II
(I) ${}^{220}\text{A} \longrightarrow {}^{100}\text{B} + {}^{120}\text{C}$	(P) 200
(II) ${}^{250}\text{A} \longrightarrow {}^{150}\text{B} + {}^{100}\text{C}$	(Q) 240
(III) ${}^{150}\text{A} \longrightarrow {}^{70}\text{B} + {}^{80}\text{C}$	(R) 80
(IV) ${}^{200}\text{A} \longrightarrow {}^{50}\text{B} + {}^{150}\text{C}$	(S) 160
	(T) 140
	(U) 25

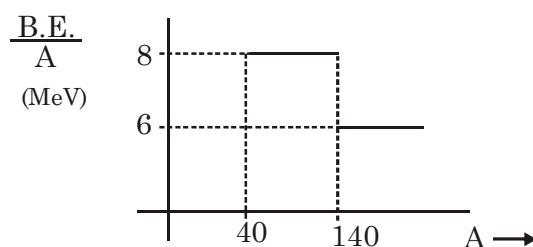
Correct match for kE of particle 'B'

- | | |
|---|--|
| (A) II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S | (B) I $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U |
| (C) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S | (D) I $\rightarrow$ Q, II $\rightarrow$ U, IV $\rightarrow$ T |

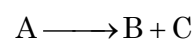
Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Hypothetical B.E. per nucleon versus mass number is given



Consider nuclear Fission reaction.



List I gives different reaction and List II given energy in MeV.

List-I	List-II
(I) $^{220}\text{A} \longrightarrow ^{100}\text{B} + ^{120}\text{C}$	(P) 200
(II) $^{250}\text{A} \longrightarrow ^{150}\text{B} + ^{100}\text{C}$	(Q) 240
(III) $^{150}\text{A} \longrightarrow ^{70}\text{B} + ^{80}\text{C}$	(R) 80
(IV) $^{200}\text{A} \longrightarrow ^{50}\text{B} + ^{150}\text{C}$	(S) 160
	(T) 140
	(U) 25

Correct match for kE of particle 'C' is

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T
 (B) II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ S
 (C) I $\rightarrow$ U, II $\rightarrow$ Q, IV $\rightarrow$ S
 (D) I $\rightarrow$ P, III $\rightarrow$ T, IV $\rightarrow$ U

Space for Rough Work

SECTION-II : (Maximum Marks: 30)

- This section contains **TEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

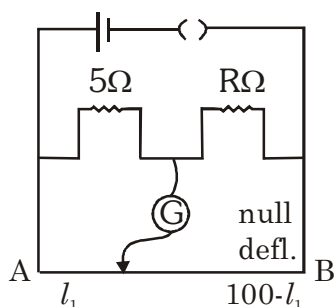
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+	●				
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

●	●	●	0	0	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

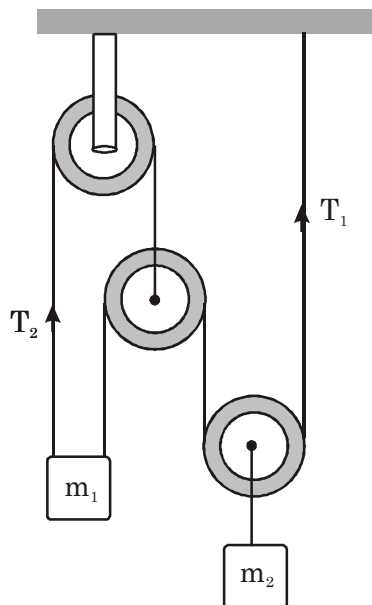
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- The resistances in the two arms of the meter bridge are 5Ω and $R\Omega$, respectively. When the resistance R is shunted with an equal resistance, the new balance point is at $1.6 l_1$. The resistance R (in Ω) is (where l_1 is the previous balance point) :-



Space for Rough Work

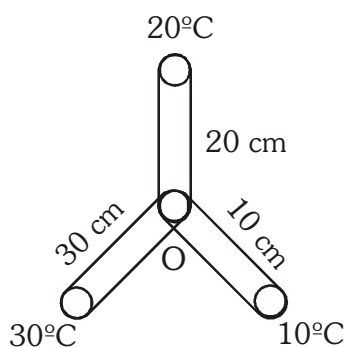
2. In a series LCR ac circuit value of R , L and C are variable and numerical value of R is always equal to numerical value of L . For a given a.c. voltage source resonant current is obtained be 1.21 A. Now capacity is increased to four times. Then new resonant current (in ampere) for the same source will be -
3. Two blocks of masses $m_1 = 5$ kg and $m_2 = 2$ kg are connected by threads which pass over the pulleys as shown in the figure. The threads are massless and the pulleys are massless and smooth. The blocks can move only along the vertical direction. If the acceleration of m_1 , if expressed in simplest form, is equal to $\left(\frac{2n}{n-1}\right) \text{ m/sec}^2$, calculate 'n'. (Take $g = 10 \text{ m/s}^2$)



Space for Rough Work

ALLEN

4. A and B are two points on horizontal ground separated by a distance of 300 m. A point source of sound is situated at point A. If velocity of sound in still air is 260 m/s and air flows with speed 100 m/s horizontally and perpendicular to AB then find time taken (in sec.) by sound to travel from A to B.
5. A particle of mass (m) is revolving around a planet of mass 'M' in circular orbit of radius '3R'. It is given $\frac{GMm}{10aR}$ energy, so that now it moves in elliptical orbit of semi major axis '5R'. Find 'a'.
6. Three rods of same material, same area of cross-section but different lengths 10 cm, 20 cm and 30 cm are connected at a point as shown. What is temperature of junction O (in °C) ?

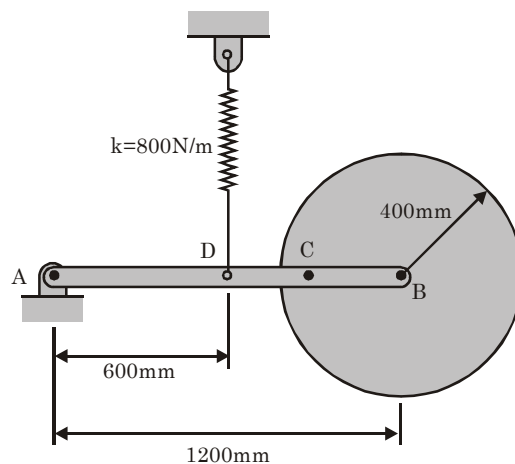


7. A hemispherical bowl of radius $R = 0.1\text{m}$ is rotating about its own axis (which is vertical) with an angular velocity ω . A particle of mass 10^{-2} kg on the frictionless inner surface of the bowl is also rotating with the same ω . The particle is at a height h from the bottom of the bowl. It is desired to measure 'g' using this set up, by measuring h accurately. Assuming that R and ω are known precisely and that the least-count in the measurement of h is 1 mm, the numerical value of minimum possible error Δg in the measured value of g is given by $\alpha\text{ m/s}^2$. Fill the value of α in OMR sheet.

Space for Rough Work

ALLEN

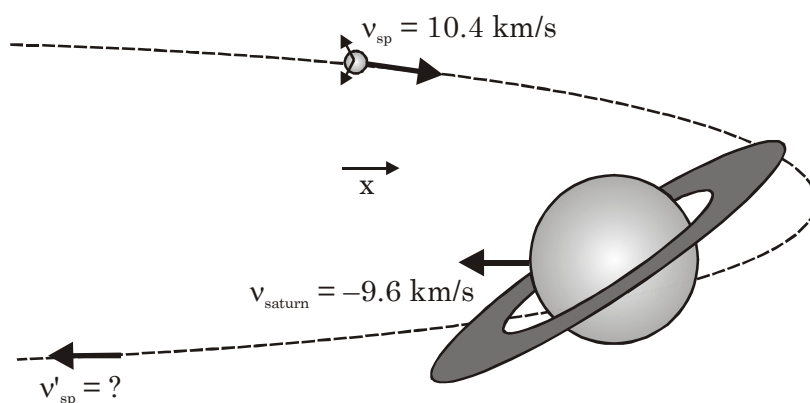
8. An ideal gas follows the relation $PV^2 = \text{constant}$. Initial pressure and volume of the gas is P_0 and V_0 respectively. Work done by the gas in increasing the volume to $2V_0$ is $\frac{2.34P_0V_0}{x}$. Find x .
9. An 8-kg uniform rod AB is hinged to a fixed support at A and is attached by means of pins B and C to a 12-kg disk of radius 400 mm. A spring attached at D holds the rod at rest in the position shown. If Point B is moved down 25 mm and released, determine the period (in sec) of vibration.



Space for Rough Work

ALLEN

10. The gravitational slingshot effect. Figure shows the planet Saturn moving in the negative x direction at its orbital speed (with respect to the Sun) of 9.6 km/s . The mass of Saturn is $5.69 \times 10^{26} \text{ kg}$. A spacecraft with mass 825 kg approaches Saturn. When far from Saturn, it moves in the $+x$ direction at 10.4 km/s . The gravitational attraction of Saturn (a conservative force) acting on the spacecraft causes it to swing around the planet (orbit shown as dashed line) and head off in the opposite direction. Estimate the final speed (in km/s) of the spacecraft after it is far enough away to be considered free of Saturn's gravitational pull.



Space for Rough Work

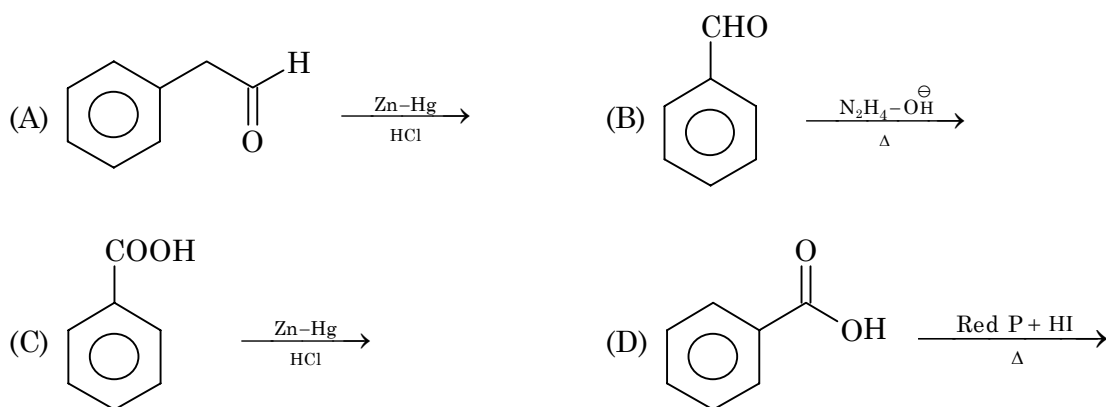
PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Zero Marks : 0 In all other cases.

1. The nature of charge on colloidal particles can be experimentally determined by which of the following process ?
 (A) Brownian movement (B) Electrophoresis
 (C) Electro-osmosis (D) Ultra microscope
2. Which of the following is/are true regarding FCC unit cell ?
 (a = edge length of unit cell, r = radius of lattice point)
 (A) The smallest distance between an octahedral void and tetrahedral void is $\frac{\sqrt{3}}{4}a$.
 (B) The smallest distance between two octahedral voids is $\frac{a}{\sqrt{2}}$.
 (C) The smallest distance between two tetrahedral voids is $\frac{\sqrt{3}}{2}a$.
 (D) The smallest distance between planes of two successive layers in ABC ABC ___type packing is $\sqrt{\frac{8}{3}}r$
3. Which of the following statements are correct about the structure of starch :
 (A) It is a polymer of α -D (+) glucose linked through α -glycosidic bond.
 (B) It consists of water soluble long unbranched chain of amylose.
 (C) It consists of water insoluble branched amylopectin having $C_1 - C_4$ and $C_1 - C_6$ glycosidic linkage between α -D glucose.
 (D) In animal body it is stored in the form of glycogen, also known as animal starch.

Space for Rough Work

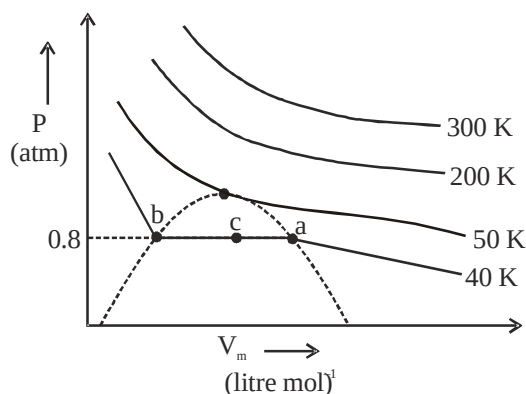
4. Identify the reactions which will yield aromatic hydrocarbons as product ?



5. Which of the following metals have total 13 electrons with magnetic quantum number value zero in the electronic configuration ?
 (A) Zn (B) Cu (C) Ca (D) Mn
6. Which of the following pairs correctly represent the comparison of their dipole moment ?
 (A) $\text{CH}_3\text{Cl} > \text{CH}_3\text{F}$ (B) $\text{CH}_3\text{Cl} > \text{CHCl}_3$
 (C) $\text{NH}_3 > \text{NF}_3$ (D) $\text{HF} > \text{HCl}$

Space for Rough Work

7. For a real gas, Andrew's isotherm is given as



Select the correct options :

- (A) At 40 K, vapour pressure of liquified gas is 0.8 atm.
 - (B) At 40 K, from point 'a' to 'b' no. of moles of liquid will increase in the mixture.
 - (C) At 35 K and 0.8 atm, gas exist in liquid phase only.
 - (D) At 200 K, gas can't be liquified.
8. Hydrolysis of which of the following will **NOT** give halogen acid (HX) as a final product ?
- (A) BF_3 (B) SiCl_4 (C) SiF_4 (D) NCl_3

Space for Rough Work

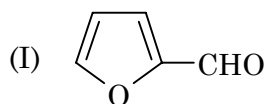
SECTION-I(ii) : (Maximum Marks : 12)

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- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
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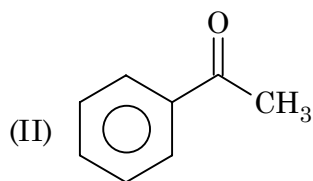
Answer the following by appropriately matching the lists based on the information given in the paragraph.

9. List-I

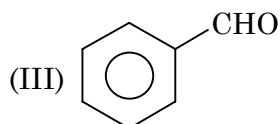
List-II



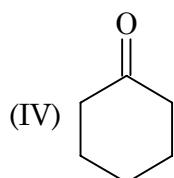
(P) Aldol condensation



(Q) Cannizzaro's reaction



(R) Positive Tollen's Test



(S) Perkins condensation

(T) Iodoform Test

(U) Benzoin condensation

Which of the following options is the only **CORRECT** combination ?

- (A) (I) - (Q) (R), (S), (U)
(C) (I) - (P), (R)

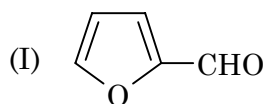
- (B) (II) - (P), (S), (T), (U)
(D) (II) - (P), (S)

Space for Rough Work

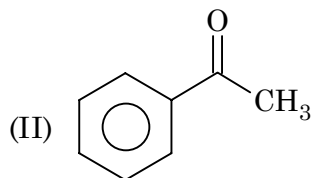
10.

List-I

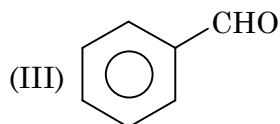
List-II



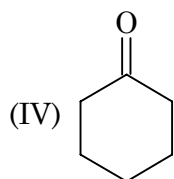
(P) Aldol condensation



(Q) Cannizzaro's reaction



(R) Positive Tollen's Test



(S) Perkins condensation

(T) Iodoform Test

(U) Benzoin condensation

Which of the following options is the only **CORRECT** combination ?

(A) (III) - (Q) (T)

(B) (III) - (Q) (R) (S) (U)

(C) (IV) - (P) (S) (U)

(D) (IV) - (P) (U) (R)

Space for Rough Work

11. List-I has four entries. The temperature is increased for I and II and the solution is diluted to two times for III and IV. List II has six entries enlisting the changes that might have occurred.

List-I

- (I) Pure water
 (II) $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
 (III) 0.1 M aqueous solution of H_2S
 ($\text{K}_{\text{a}_1} = 10^{-7}$, $\text{K}_{\text{a}_2} = 10^{-14}$)
 (IV) 0.1 M aqueous solution of NaHCO_3
 (K_{a_1} of $\text{H}_2\text{CO}_3 = 10^{-7}$ and $\text{K}_{\text{a}_2} = 10^{-11}$)

List-II

- (P) pH decreases
 (Q) degree of ionisation increases
 (R) $[\text{HS}^-]$ stays at 10^{-14} M
 (S) Equilibrium constant changes
 (T) Reaction goes backward
 (U) Solution stays basic

Which of the following combination is **CORRECT**?

- (A) (I) (P) (Q) (S) (T) (B) (I) (P) (Q) (S)
 (C) (II) (Q) (T) (D) (II) (S) (T)

Space for Rough Work

12. List-I has four entries. The temperature is increased for I and II and the solution is diluted to two times for III and IV. List II has six entries enlisting the changes that might have occurred.

List-I

- (I) Pure water
 (II) $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
 (III) 0.1 M aqueous solution of H_2S
 ($\text{K}_{\text{a}_1} = 10^{-7}$, $\text{K}_{\text{a}_2} = 10^{-14}$)
 (IV) 0.1 M aqueous solution of NaHCO_3
 (K_{a_1} of $\text{H}_2\text{CO}_3 = 10^{-7}$ and $\text{K}_{\text{a}_2} = 10^{-11}$)

List-II

- (P) pH decreases
 (Q) degree of ionisation increases
 (R) $[\text{HS}^-]$ stays at 10^{-14} M
 (S) Equilibrium constant changes
 (T) Reaction goes backward
 (U) Solution stays basic

Which of the following combination is **CORRECT**?

- (A) (III) (P) (Q) (R) (B) (III) (P) (Q) (R) (T)
 (C) (IV) (P) (Q) (D) (IV) (U)

Space for Rough Work

SECTION-II : (Maximum Marks: 30)

- This section contains **TEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

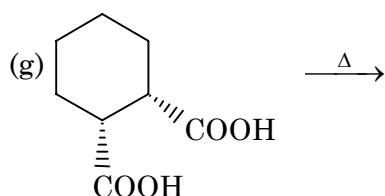
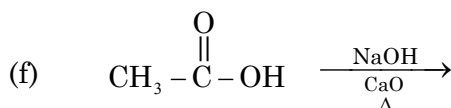
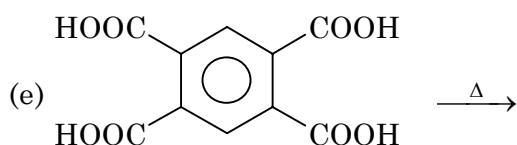
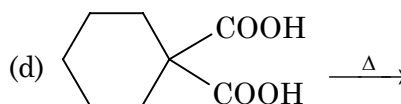
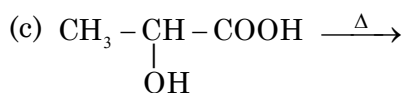
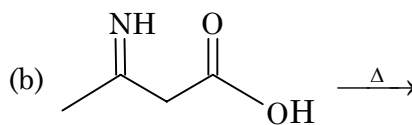
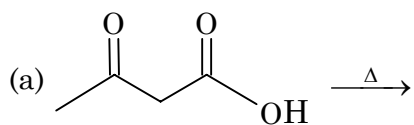
-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

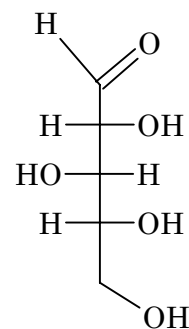
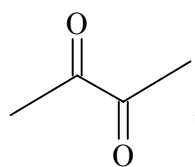
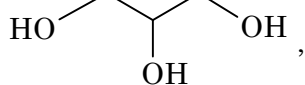
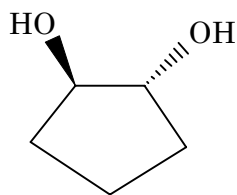
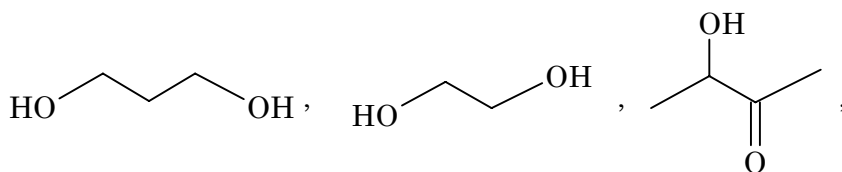
1. $\text{KI}_{(\text{aq.})}$ and $\text{KIO}_{3(\text{aq.})}$ mixed together in presence of HCl to form I_2 and KCl and liberated I_2 required 100 mL of 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ solution for complete reduction of I_2 .
 If x = millimoles of KI used in the reaction.
 and y = millimoles of HCl used in the reaction.
 then value of $(x + y)$ is :
2. A sample contains H-atoms, He^+ ions, Li^{2+} ions with H-atoms being in 2nd state, He^+ ions in 4th state and Li^{2+} ions in 6th state. Find the different no. of spectral lines possible from the sample on de-excitation of all the atoms and ions to ground state.

Space for Rough Work

3. Total number of reactions involves decarboxylation under specified condition.

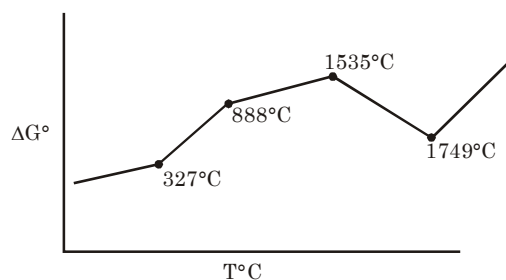


4. How many of the following compounds, will undergo oxidative cleavage with HIO_4 ?



Space for Rough Work

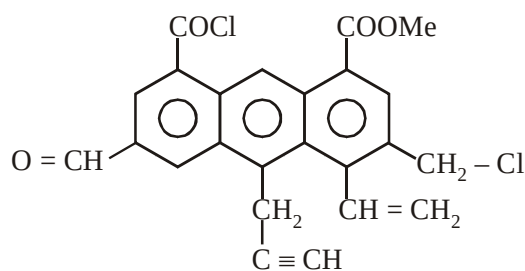
5. Total number of meso compounds possible for $C_7H_{14}Cl_2$?
6. It is the plot of ΔG° vs T for $Pb \rightarrow PbO$



Find the sum of values of boiling point of Pb and melting point of PbO in K.

Space for Rough Work

7. No. of neutral oxides formed by nitrogen is :
8. Which of the following metals forms MO_2 type compound as a major product when it burns with excess of oxygen gas?
Li, Na, Mg, K, Rb, Cs, C, Be, S, Al
9. Find the total number optically active isomer possible for $[\text{Pt}(\text{NH}_2\text{--CH}(\text{CH}_3)\text{--CO--O})_2]^\circ$.
10. 1 mol of following compound can consume maximum reaction many mole of MeMgBr ?



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Zero Marks : 0 In all other cases.

1. Suppose $f(x)$ be a twice differentiable polynomial function such that $f(1) = 2$, $f(2) = 8$ and $f(3) = 18$, then

- (A) $f''(x) = 4 \quad \forall x \in R$
 (B) there exist at least one $x \in (1, 3)$ such that $f''(x) = 4$.
 (C) there exist at least one $x \in (2, 3)$ such that $f'(x) - 10 = 0$.
 (D) there exist at least one $x \in (1, 3)$ such that $f'(x) - 8 = 0$.

2. Two perpendicular unit vectors $\vec{a}$ and $\vec{b}$ are such that $[\vec{r} \vec{a} \vec{b}] = \frac{5}{4}$, $\vec{r} \cdot (3\vec{a} + 2\vec{b}) = 0$ and

$$\int_{-2\vec{r} \cdot \vec{a}}^{\frac{-4}{3}\vec{r} \cdot \vec{b}} \frac{x+1}{x^2+1} dx = \frac{\pi}{2}. \text{ Which of the following is (are) correct ?}$$

- (A) $|\vec{r}|^2 = \frac{19}{8}$ (B) $|\vec{r}| = \frac{7}{4}$
 (C) $\vec{r} = \frac{\vec{a}}{2} - \frac{3\vec{b}}{4} + \frac{5}{4}(\vec{a} \times \vec{b})$ (D) $\vec{r} = \frac{-\vec{a}}{2} + \frac{3\vec{b}}{2} + \frac{5}{4}(\vec{a} \times \vec{b})$

Space for Rough Work

3. The sum of real roots of the equation $(2x - 1)^2 - 3|2x - 1| + 2 = 0$ is -
 (A) 3 (B) 2 (C) 1 (D) 0
4. Which of the following definite integral is equal to zero
 (A) $\int_{-1}^1 (\tan^3 x + \sin^5 x) dx$ (B) $\int_0^{\pi/2} \ln(\tan x) dx$
 (C) $\int_{-1}^1 (x^3 + x^7) dx$ (D) $\int_{-\pi/2}^{\pi/2} \sin^2 x dx$
5. z is a variable complex number satisfying $|z+1| - |z-1| = \frac{3}{2}$ then identify the incorrect statements
 (A) locus of z is a circle (B) locus of z is a ellipse
 (C) the maximum value of $|z|$ is $\frac{3}{4}$ (D) the least value of $|z|$ is $\frac{3}{4}$

Space for Rough Work

6. Let $A_0, A_1, A_2, A_3, A_4, A_5$ are the consecutive vertices of a regular hexagon inscribed in a circle of unit radius. Which of the following holds 'TRUE'.
- (A) Product of lengths of segments A_0A_1, A_0A_2, A_0A_4 is 3
- (B) Area of hexagon $A_0A_1A_2A_3A_4A_5$ is $3\sqrt{3}$
- (C) Area of triangle $A_0A_1A_2$ is $\frac{\sqrt{3}}{4}$
- (D) Area of quadrilateral $A_0A_2A_3A_5$ is $\sqrt{3}$
7. Given A and C are involutory matrices and B is a non-singular matrix, then always $(AB^{-1}C)^{-1}$ is equal to -
- (A) $A^{-1}BC^{-1}$ (B) ABC (C) ABC^{-1} (D) CBA
8. The probability that a man can hit a target is $\frac{3}{4}$. He tries 5 times. The probability that he will
- (A) hit the target atleast 3 times is $\frac{459}{512}$ (B) hit the target exactly once is $\frac{15}{1024}$
- (C) miss the target all 5 times is $\frac{1}{1024}$ (D) hit the target atmost two times in $\frac{45}{512}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Four or Five or Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $L(n, r)$ denotes the number of ways to select 'r' objects from 'n' distinct objects placed in a row, such that no two of the selected objects were consecutive, and let $R(n, r)$ denotes the number of ways to select 'r' objects from 'n' distinct objects placed on a circle, such that no two of the selected objects were consecutive.

List-I

- (I) $R(12, 4) + L(8, 2) =$
 (II) $R(14, 5) + L(10, 3) =$
 (III) $L(12, 4) =$
 (IV) $L(13, 5) =$

List-II

- (P) ${}^{10}C_5$
 (Q) ${}^8C_4 + {}^8C_3$
 (R) 9C_5
 (S) 8C_4
 (T) ${}^{11}C_5 - {}^{10}C_4$

Which of the following is / are correct :

- (A) $I \rightarrow P$ (B) $II \rightarrow P, T$ (C) $III \rightarrow P, Q, R$ (D) $IV \rightarrow S$

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $L(n, r)$ denotes the number of ways to select 'r' objects from 'n' distinct objects placed in a row, such that no two of the selected objects were consecutive, and let $R(n, r)$ denotes the number of ways to select 'r' objects from 'n' distinct objects placed on a circle, such that no two of the selected objects were consecutive.

List-I

- (I) $R(12, 4) + L(8, 2) =$
 (II) $R(14, 5) + L(10, 3) =$
 (III) $L(12, 4) =$
 (IV) $L(13, 5) =$

List-II

- (P) ${}^{10}C_5$
 (Q) ${}^8C_4 + {}^8C_3$
 (R) 9C_5
 (S) 8C_4
 (T) ${}^{11}C_5 - {}^{10}C_4$

Which of the following is / are correct :

- (A) $I \rightarrow Q, R$ (B) $II \rightarrow Q, R$ (C) $III \rightarrow S$ (D) $IV \rightarrow P, Q, R$

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

$P(\alpha_1, \alpha_2, \alpha_3)$ is the image of $(2, 1, 4)$ in the plane $2x - y + z + 5 = 0$ and $Q(\beta_1, \beta_2, \beta_3)$ is the foot of perpendicular from $(2, 1, 4)$ in the plane $2x - y + z + 5 = 0$, then

List-I

- (I) $\alpha_1 + \alpha_2 + \alpha_3$
 (II) $\alpha_2 + \alpha_3$
 (III) $\alpha_1 + \beta_1$
 (IV) $\beta_1 + \beta_2$

List-II

- (P) -1
 (Q) 5
 (R) 3
 (S) -8
 (T) 1

Which of the following is correct?

- (A) I-P, III -R (B) I-P, II-T (C) II-R, IV-S (D) I-P, II-Q, III-S

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

P $(\alpha_1, \alpha_2, \alpha_3)$ is the image of $(2, 1, 4)$ in the plane $2x - y + z + 5 = 0$ and Q $(\beta_1, \beta_2, \beta_3)$ is the foot of perpendicular from $(2, 1, 4)$ in the plane $2x - y + z + 5 = 0$, then

List-I

(I) $\alpha_1 + \alpha_2 + \alpha_3$

(II) $\alpha_2 + \alpha_3$

(III) $\alpha_1 + \beta_1$

(IV) $\beta_1 + \beta_2$

List-II

(P) -1

(Q) 5

(R) 3

(S) -8

(T) 1

Which of the following is correct?

(A) III-S

(B) II-R

(C) III-T

(D) IV-T, II-S

Space for Rough Work

SECTION-II : (Maximum Marks: 30)

- This section contains **TEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If the equation $3x^2 - 10xy + 3y^2 = 0$ represents adjacent sides of rhombus and one of the diagonal is $x + y = 4$. If the vertices of the rhombus are $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ & (x_4, y_4) .
Find $\frac{x_1^2 + x_2^2 + x_3^2 + x_4^2}{y_1^2 + y_2^2 + y_3^2 + y_4^2}$
2. Let in a non-constant Arithmetic progression of 'n' terms, 'a' be the first term, 'b' be the last term. 'c' be sum of all the terms and 'd' be the common difference then find x, where
 $d = \frac{a^2 - b^2}{a + b - cx}$. (Where $c \neq 0, n \in \mathbb{N}$)
3. If a, b, c are positive real numbers and the system of equations $(a-1)x = y + z, (b-1)y = z + x$ and $(c-1)z = x + y$ has a non trivial solution, then find the minimum value of $\frac{abc}{10}$.

Space for Rough Work

4. Two perpendicular tangents are drawn from point P to a circle of radius 5, which touches the circle at A & B. At a point M on arc AB a tangent is drawn to circle which intersect PA & PB at Q and R respectively, then perimeter of ΔPQR is

5. P (4, 2) is a point on parabola $x^2 = 8y$. A tangent and a normal are drawn at point P which

intersects y axis at point T & N respectively, then value of $\frac{1}{10} \left(\frac{PN}{\sin(\angle PTN)} \right) =$

6. Let $\left(2 + \frac{x^4}{x^2 + 1} \right)^{10} = A_{20}x^{20} + A_{19}x^{19} + A_{18}x^{18} + \dots + A_1x + A_0 + \frac{B_1}{x^2 + 1} + \frac{B_2}{(x^2 + 1)^2} + \dots + \frac{B_{10}}{(x^2 + 1)^{10}}$ where

$A_{20}, A_{19}, A_{18}, \dots, A_1, A_0, B_1, B_2, \dots, B_{10}$ are constants then find A_0 .

Space for Rough Work

7. Number of integral values of λ for which radius r of circle $(\alpha^2 - \alpha)x^2 + 2(\alpha + 2)y^2 + (\alpha^2 - 1)xy + 12\alpha x + (\alpha + 9)y + 4\lambda = 0$, ($\alpha \in \mathbb{R}$) is such that $r \in (3, 5)$ is
8. If $\lim_{x \rightarrow 0} \frac{\left(1 - \cos\left(1 - \cos\frac{x}{2}\right)\right)}{2^m x^n}$ is equal to the lefthand derivative of $e^{-|x|}$ at $x = '0'$. Then find the value of $\left(\frac{n-10m}{10}\right)$.
9. If $x = \sin\left(\frac{2\pi}{7}\right) + \sin\left(\frac{3\pi}{7}\right) - \sin\left(\frac{\pi}{7}\right)$, $y = \sin\left(\frac{\pi}{14}\right)\sin\left(\frac{3\pi}{14}\right)\sin\left(\frac{5\pi}{14}\right)$, $z = \sin\left(\frac{\pi}{7}\right)\sin\left(\frac{2\pi}{7}\right)\sin\left(\frac{3\pi}{7}\right)$, then the value of $\left(\frac{z}{xy}\right)$ is
10. In $\mathbb{R}^2$, let $A(1, 1)$, $B(2, 1)$, $C(5, 1)$ and $D(\alpha, \beta^2 + 1)$ be such that triangle ACD is an equilateral triangle. If C_1 and C_2 are incircle of $\triangle ABD$ and $\triangle BCD$ then find $[r_1 \ r_2]$. Where r_1 and r_2 are radius of C_1 and C_2 respectively and $[.]$ denotes greatest integer function.

Space for Rough Work